

GTEx, PrediXcan, TWAS

HUGEN 2073

Jon Chernus

Motivation

So far, we've mainly discussed “variant-levels” methods/tools

- Annotating (GWAS) variants
- Finding evidence a (GWAS) variant may be involved in a trait (primarily by “mapping” it to a gene)

Motivation

We've also discussed some gene-based methods/tools

- Gene-based association tests (e.g., MAGMA)
- Finding evidence a pathway/functional category may be involved in a trait (pathway/gene set analysis)

Motivation

Often we think of genes as the “bricks” of the “genetic architecture.”

We ask, “Does this particular (GWAS) variant somehow affect a gene’s function in a way that’s relevant to a particular trait?”

If gene function takes us a step closer to the biology of a trait...

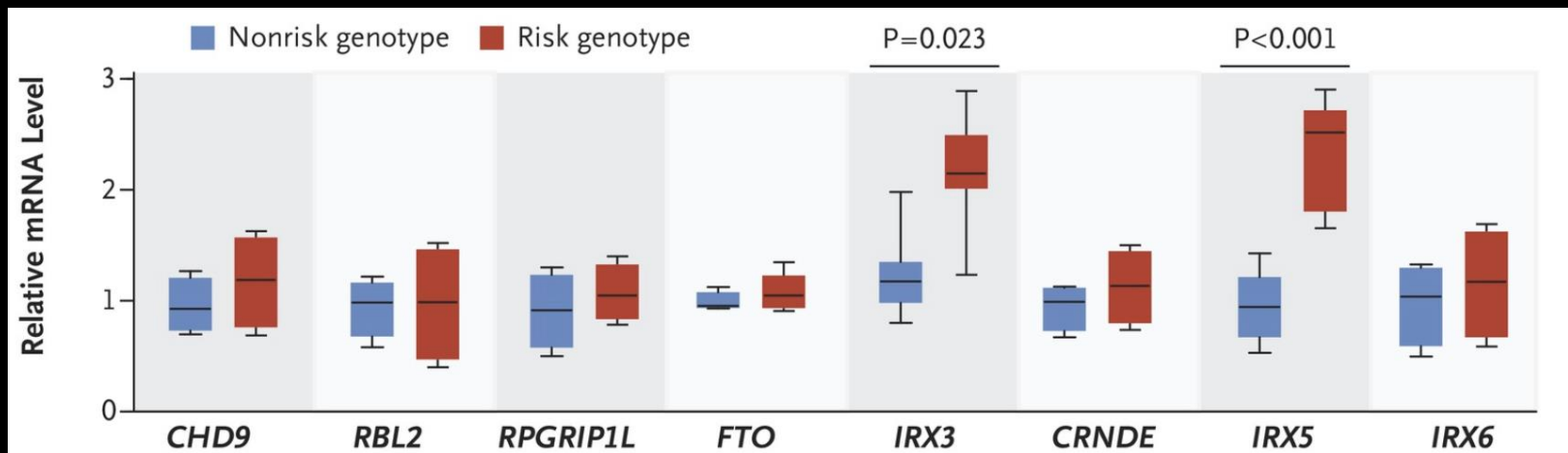
- Can we quantify/define “gene function” in a simple/systematic way?
- Can we actually measure this for individuals somehow?
- Can we then test each gene’s “gene function” for association with a trait?

Review of eQTLs

What is an eQTL?

Expression quantitative trait locus

An eQTL is a locus at which genotype affects transcription of a gene.



Why do we care about eQTLs?

???

Why we care about eQTLs

eQTLs are enriched in GWAS signals.

See, e.g.

Nicolae, D. L., Gamazon, E., Zhang, W., Duan, S., Dolan, M. E., & Cox, N. J. (2010). Trait-associated SNPs are more likely to be eQTLs: annotation to enhance discovery from GWAS. *PLoS genetics*, 6(4), e1000888.

<https://doi.org/10.1371/journal.pgen.1000888>

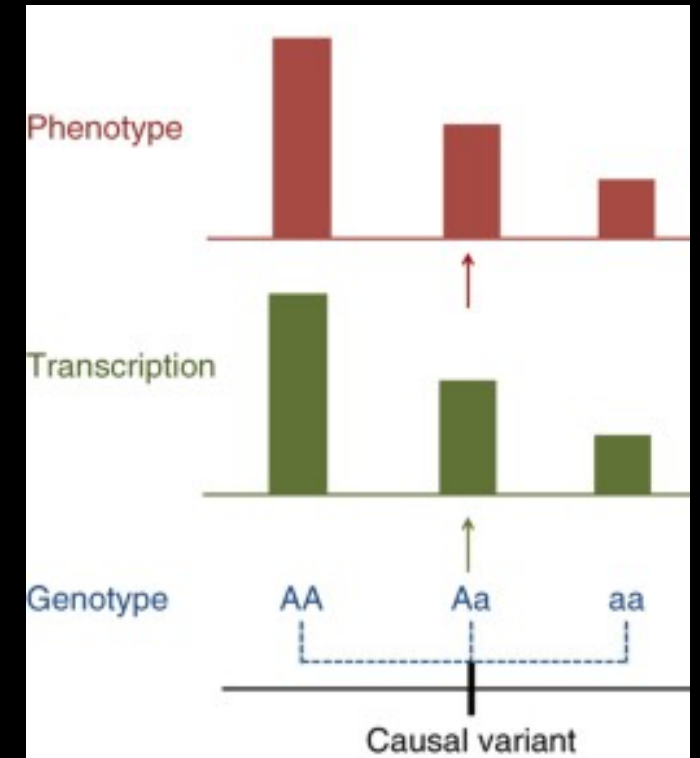
<https://pubmed.ncbi.nlm.nih.gov/20369019/>

And, if we think a gene matters for a trait, we might be interested in the effect of its expression level.

For a discussion, see

Umans, B. D., Battle, A., & Gilad, Y. (2021). Where Are the Disease-Associated eQTLs?. *Trends in genetics : TIG*, 37(2), 109–124. <https://doi.org/10.1016/j.tig.2020.08.009>

[https://www.cell.com/trends/genetics/fulltext/S0168-9525\(20\)30209-2](https://www.cell.com/trends/genetics/fulltext/S0168-9525(20)30209-2)



What are some complicating factors about eQTLs?

eQTL complications

Context is important

- variant, gene, and tissue
 - Has anyone even studied the “right” kind of cell for your trait?
- Other factors may matter, too
 - Environment (have relevant conditions been studied?)
 - Stimuli (ditto)
 - Population?
- Other considerations
 - Post-mortem samples
 - Cis- vs. trans-eQTLs
 - Splicing eQTLs
 - Transcription $\neq$ translation
 - Correlation $\neq$ causation
 - Rare variants may be poorly represented in databases

Where can we find transcriptome data?

Some examples of databases with gene expression:

- GTEx
- GEUVADIS (Genetic European Variation in Health and Disease)
- GEO (Gene Expression Omnibus)
- eQTLGen
- ArrayExpress
- TCGA
- Expression Atlas
- dbGaP
- Human Cell Atlas (HCA)
- Single Cell Portal (Broad Institute)
- Many others...

GTEx

<https://www.gtexportal.org/home/>

<https://www.gtexportal.org/home/documentationPage> (summary of methods)

GTEx (Genotype-Tissue Expression project) is a database of eQTLs.

GTE_x

Dataset Summary of Analysis Samples

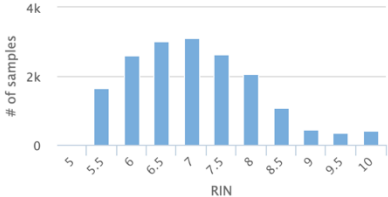
Data Source: GTEx Analysis Release V8 (dbGaP Accession phs000424.v8.p2)

V8 Release	# Tissues	# Donors	# Samples
Total	54	948	17382
With Genotype	54	838	15253
Has eQTL Analysis*	49	838	15201

* Number of samples with genotype >= 70

* Number of samples with genotype ≥ 7

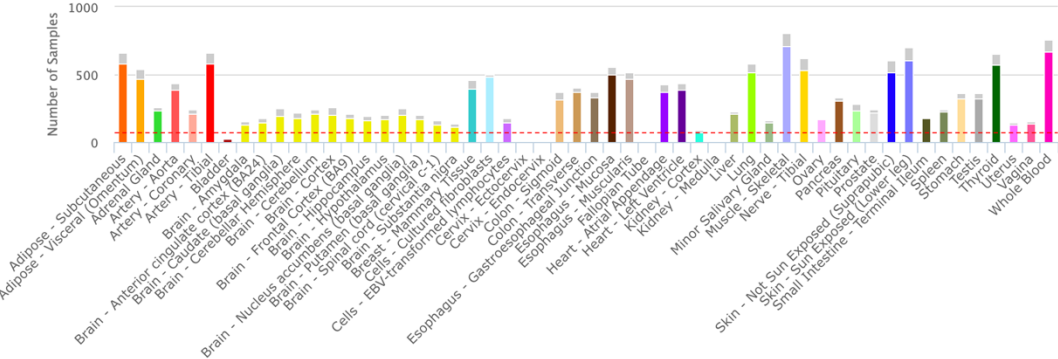
Sample RIN Histogram



V8 Sample Counts by Tissues

Sort tissues by: Tissue alphabetical

Download

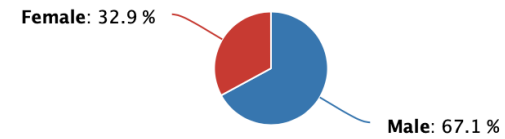


GTE_x

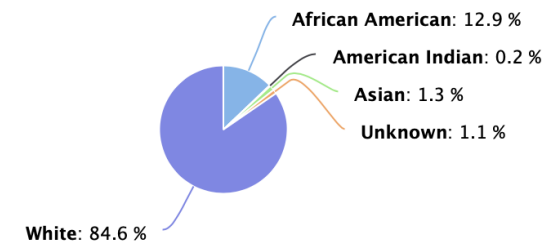
V8 Donor Info

 [Download](#)

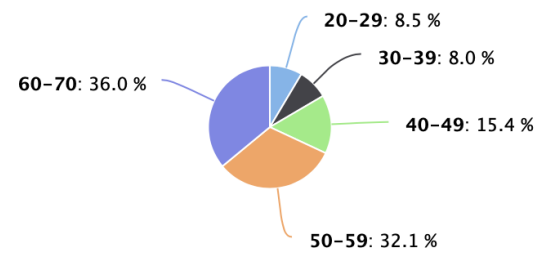
Sex



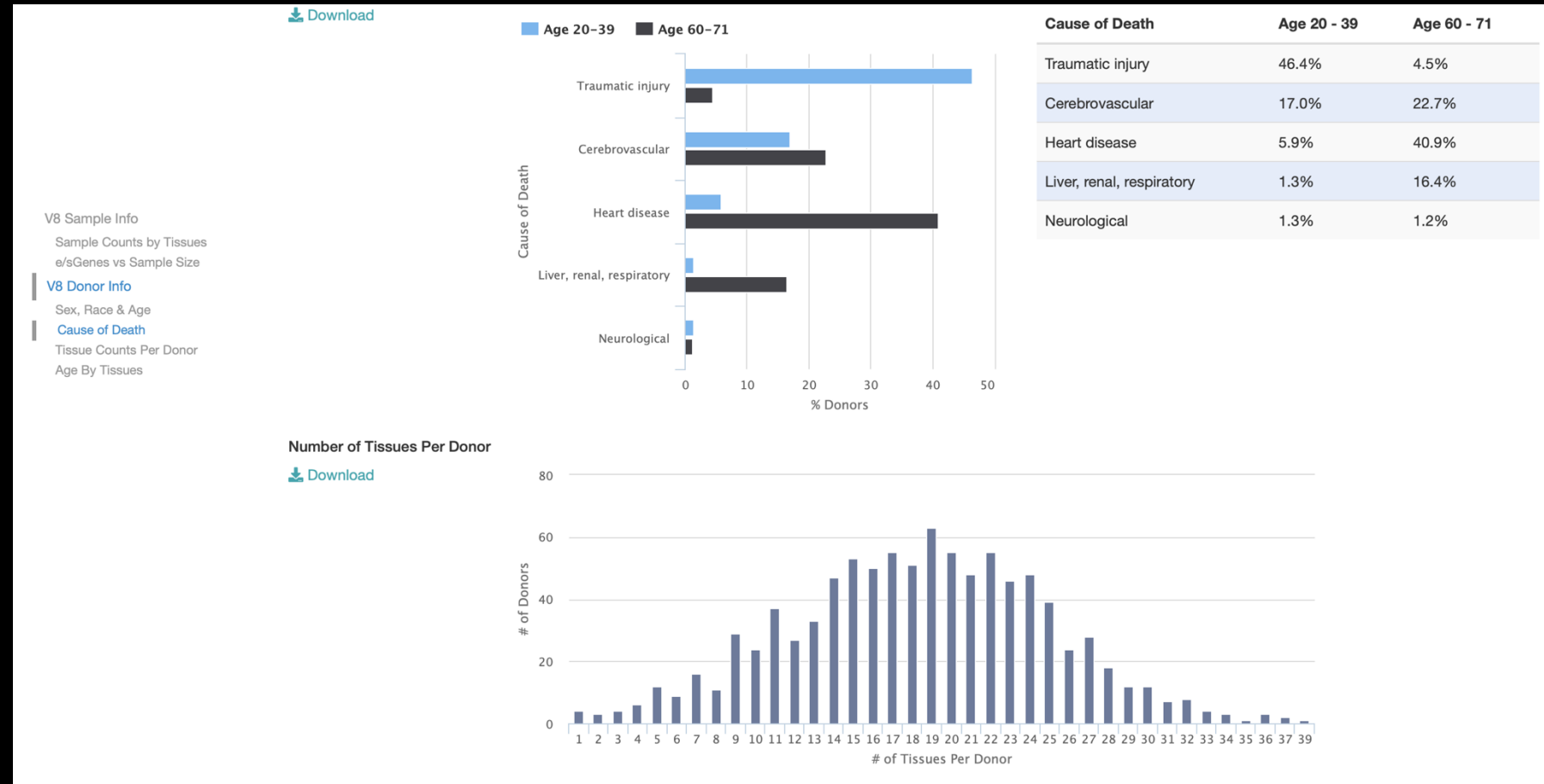
Race



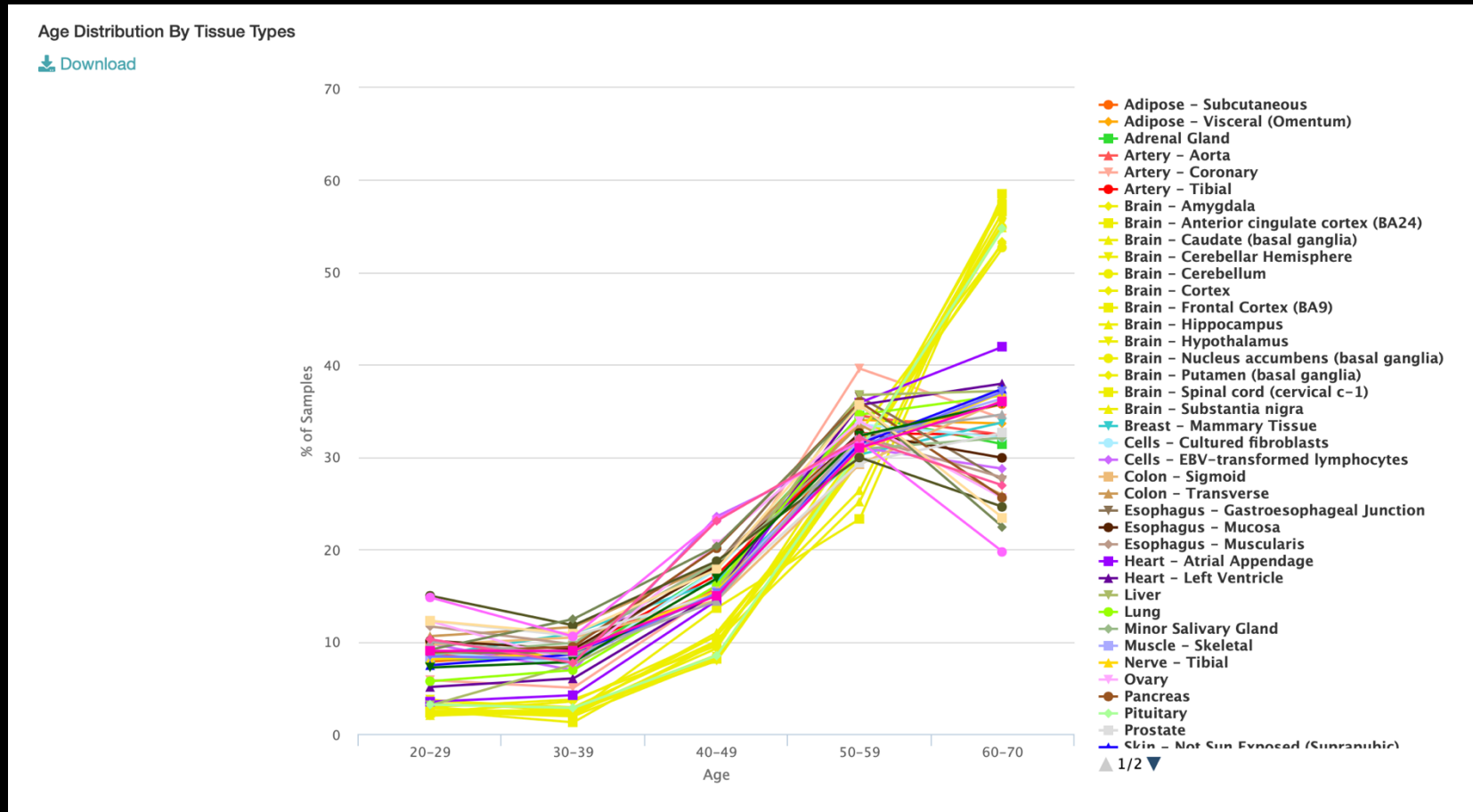
Age



GTEx



GTEx



GTE_x

[About GTEx](#)
[Publications](#)
[Access Biospecimens](#)
[FAQs](#)
[Contact](#)

[Home](#)
[Datasets](#)
[Expression](#)
[QTLs & Browser](#)
[Sample Data](#)
[Documentation](#)

[Sign In](#)

2020-01-06

RNA-seq Coverage Tracks

We have added tissue-specific RNA-seq coverage tracks to the GTEx IGV Browser. The revised track menu allows you to select the coverage tracks.

Also, we have upgraded the version of IGV and fixed bugs in the GTEx IGV Browser track menu.

Resource Overview

Current Release (V8)

[Tissue & Sample Statistics](#)
[Tissue Sampling Info \(Anatomogram\)](#)
[Access & Download Data](#)
[Release History](#)
[How to cite GTEx?](#)

The Genotype-Tissue Expression (GTEx) project is an ongoing effort to build a comprehensive public resource to study tissue-specific gene expression and regulation. Samples were collected from 54 non-diseased tissue sites across nearly 1000 individuals, primarily for molecular assays including WGS, WES, and RNA-Seq. Remaining samples are available from the GTEx Biobank. The GTEx Portal provides open access to data including gene expression, QTLs, and histology images.

Explore GTEx

Browse

[By Tissue](#)

[Histology Image Viewer](#)

Browse and search all data by gene

Browse and search all data by variant

Browse and search all data by tissue

Browse and search GTEx histology images

Expression

[Multi-Gene Query](#)

[Top 50 Expressed Genes](#)

[Transcript Browser](#)

[Single Cell Expression Data](#)

Browse and search expression by gene and tissue

Visualize the top 50 expressed genes in each tissue

Visualize transcript expression and isoform structures

Coming soon!

QTL

[Locus Browser](#)

[Locus Browser VC](#)

[IGV Browser](#)

[eQTL Dashboard](#)

[eQTL Calculator](#)

Visualize QTLs by gene in the Locus Browser

Visualize QTLs by variant in the Locus Browser VC (Variant Centric)

Visualize tissue-specific eQTLs and coverage data in the IGV Browser

Batch query eQTLs by gene and tissue

Test your own eQTLs

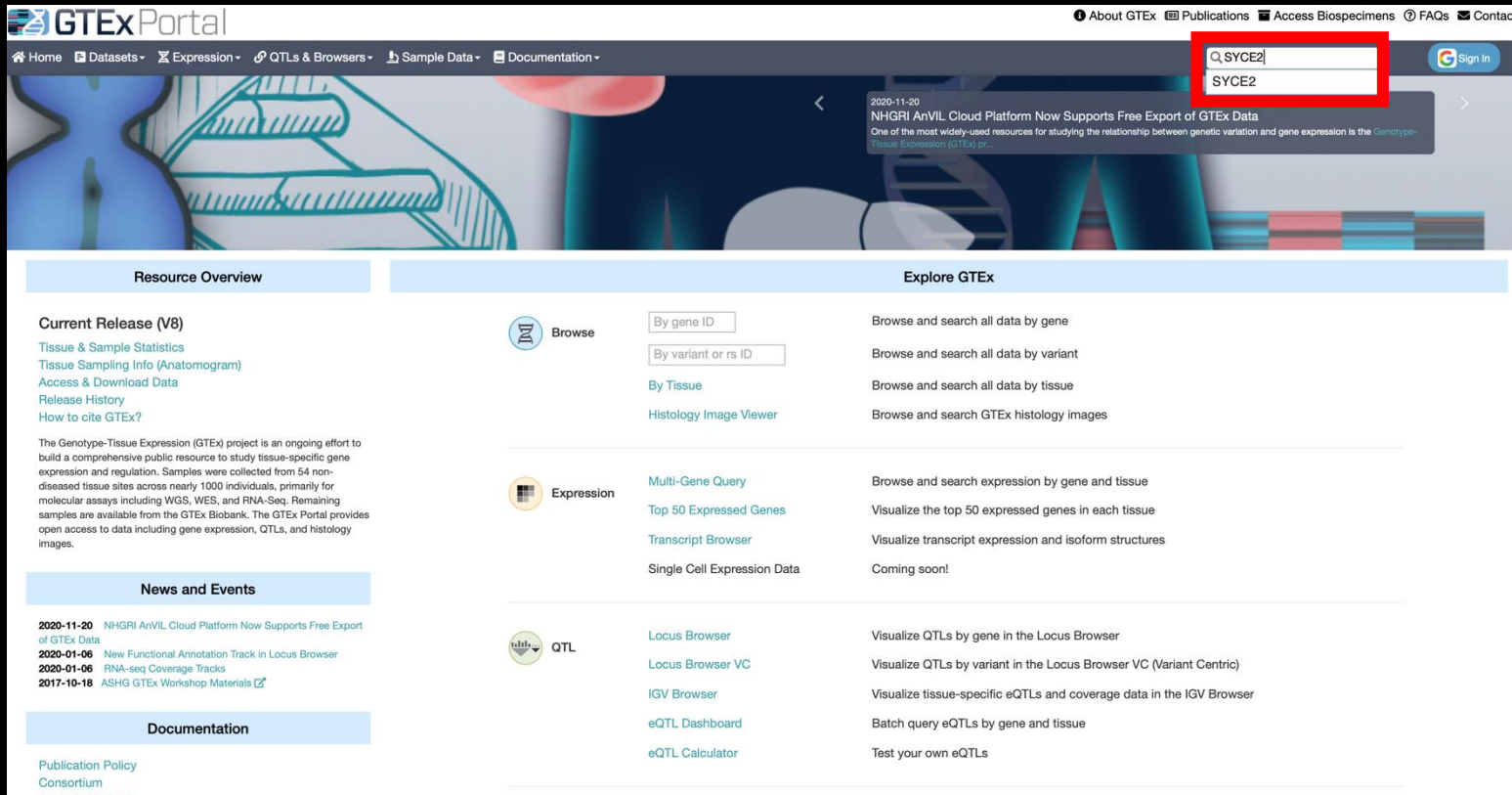
News and Events

2020-11-20 [NHGRI AnVIL Cloud Platform Now Supports Free Export of GTEx Data](#)
2020-01-06 [New Functional Annotation Track in Locus Browser](#)
2020-01-06 [RNA-seq Coverage Tracks](#)
2017-10-18 [ASHG GTEx Workshop Materials](#)

Documentation

GTEx Portal Examples

Search a gene, e.g., *SYCE2*



The screenshot shows the GTEx Portal homepage. At the top, there is a navigation bar with links: Home, Datasets, Expression, QTLs & Browser, Sample Data, and Documentation. On the right side of the navigation bar, there is a search bar with the text "SYCE2" entered, highlighted by a red box. Below the navigation bar, there is a large banner image with a blue and white background. To the right of the banner, there is a news item dated 2020-11-20: "NHGRI AnVIL Cloud Platform Now Supports Free Export of GTEx Data". Below the banner, there is a "Resource Overview" section on the left and an "Explore GTEx" section on the right. The "Resource Overview" section includes links for "Current Release (V8)", "Tissue & Sample Statistics", "Tissue Sampling Info (Anatomogram)", "Access & Download Data", "Release History", and "How to cite GTEx?". The "Explore GTEx" section is divided into three main categories: "Browse", "Expression", and "QTL". Each category has a list of tools and links. The "Browse" category includes "By gene ID", "By variant or rs ID", "By Tissue", and "Histology Image Viewer". The "Expression" category includes "Multi-Gene Query", "Top 50 Expressed Genes", "Transcript Browser", and "Single Cell Expression Data". The "QTL" category includes "Locus Browser", "Locus Browser VC", "IGV Browser", "eQTL Dashboard", and "eQTL Calculator".

GTExPortal

About GTEx Publications Access Biospecimens FAQs Contact

Home Datasets Expression QTLs & Browser Sample Data Documentation

Search SYCE2

Sign In

2020-11-20
NHGRI AnVIL Cloud Platform Now Supports Free Export of GTEx Data
One of the most widely-used resources for studying the relationship between genetic variation and gene expression is the [Genotype-Tissue Expression \(GTEx\) project](#).

Resource Overview

Current Release (V8)

- [Tissue & Sample Statistics](#)
- [Tissue Sampling Info \(Anatomogram\)](#)
- [Access & Download Data](#)
- [Release History](#)
- [How to cite GTEx?](#)

The Genotype-Tissue Expression (GTEx) project is an ongoing effort to build a comprehensive public resource to study tissue-specific gene expression and regulation. Samples were collected from 54 non-diseased tissue sites across nearly 1000 individuals, primarily for molecular assays including WGS, WES, and RNA-Seq. Remaining samples are available from the GTEx Biobank. The GTEx Portal provides open access to data including gene expression, QTLs, and histology images.

News and Events

- [2020-11-20 NHGRI AnVIL Cloud Platform Now Supports Free Export of GTEx Data](#)
- [2020-01-06 New Functional Annotation Track in Locus Browser](#)
- [2020-01-06 RNA-seq Coverage Tracks](#)
- [2017-10-18 ASHG GTEx Workshop Materials](#)

Documentation

- [Publication Policy](#)
- [Consortium](#)
- [Analysis Methods](#)

Explore GTEx

Browse

-
-
- [By Tissue](#)
- [Histology Image Viewer](#)

- Browse and search all data by gene
- Browse and search all data by variant
- Browse and search all data by tissue
- Browse and search GTEx histology images

Expression

- [Multi-Gene Query](#)
- [Top 50 Expressed Genes](#)
- [Transcript Browser](#)
- [Single Cell Expression Data](#)

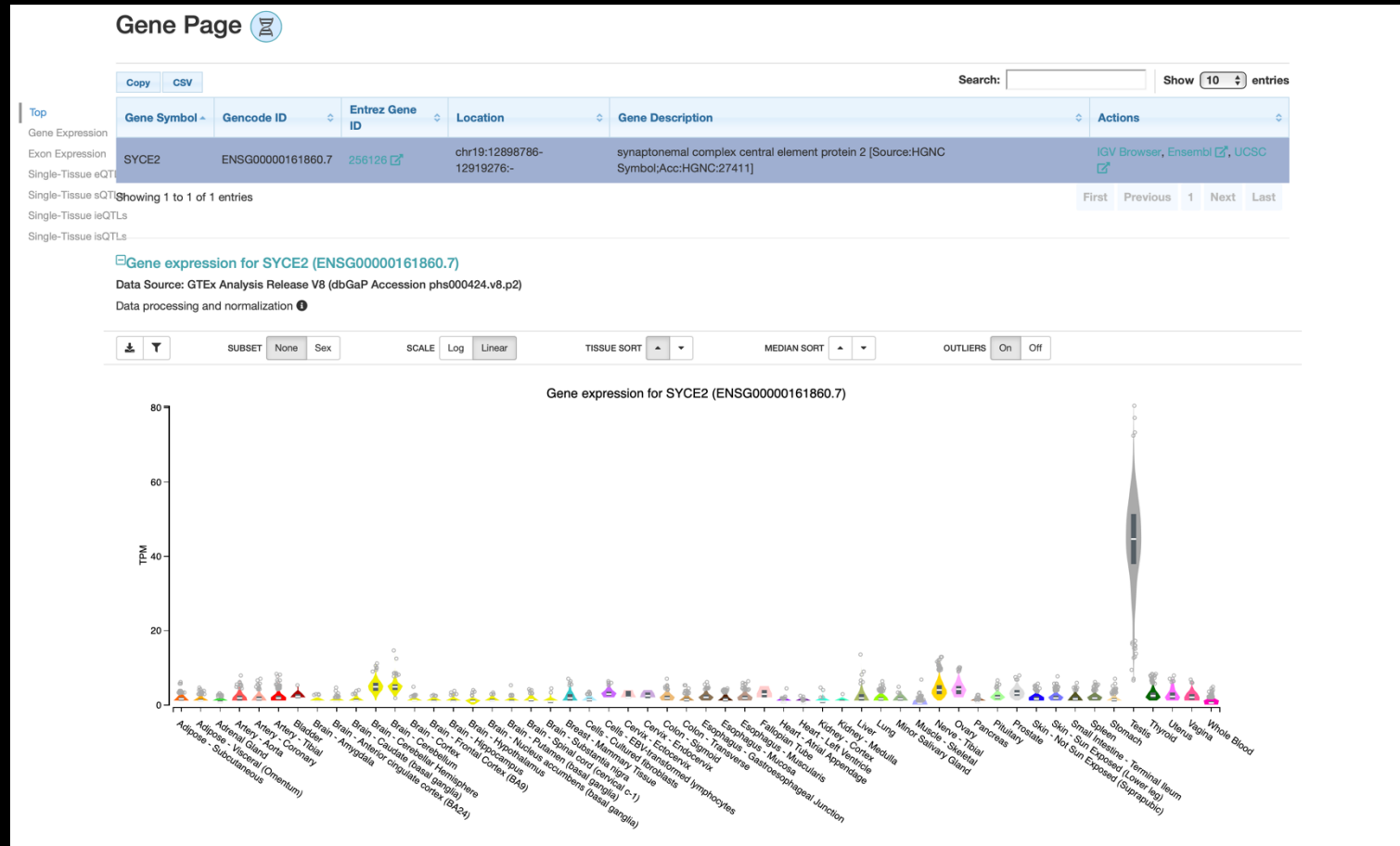
- Browse and search expression by gene and tissue
- Visualize the top 50 expressed genes in each tissue
- Visualize transcript expression and isoform structures
- Coming soon!

QTL

- [Locus Browser](#)
- [Locus Browser VC](#)
- [IGV Browser](#)
- [eQTL Dashboard](#)
- [eQTL Calculator](#)

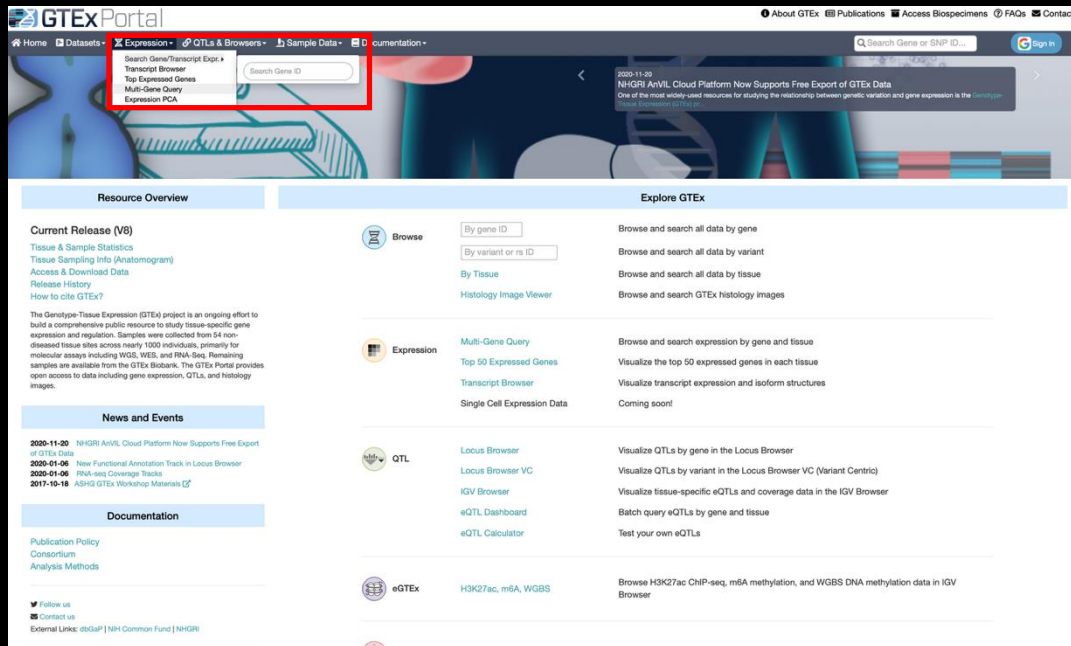
- Visualize QTLs by gene in the Locus Browser
- Visualize QTLs by variant in the Locus Browser VC (Variant Centric)
- Visualize tissue-specific eQTLs and coverage data in the IGV Browser
- Batch query eQTLs by gene and tissue
- Test your own eQTLs

GTEx Portal Examples



GTEx Portal Examples

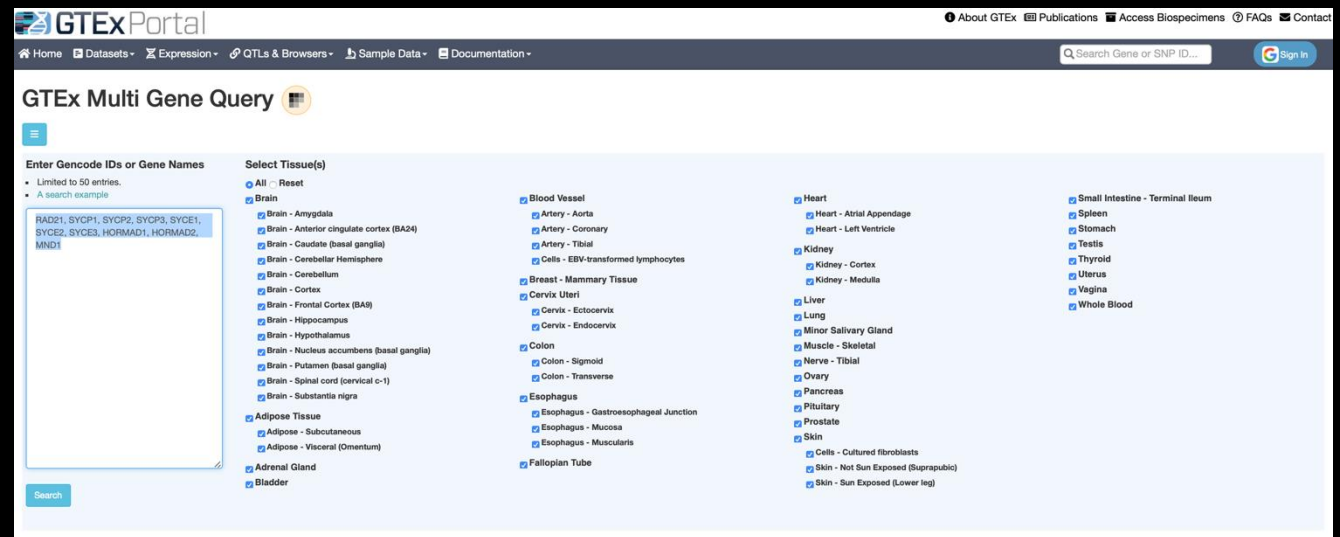
Search multiple genes



The screenshot shows the GTEx Portal homepage. A red box highlights the search bar and a dropdown menu. The dropdown menu contains the following options:

- Search Gene/Transcript Expression Browser
- Top Expressed Genes
- Multi-Gene Query
- Expression PCA

The search bar contains the text "Search Gene ID".



The screenshot shows the GTEx Multi Gene Query page. The page is titled "GTEx Multi Gene Query" and features a search bar and a list of tissues. The search bar contains the text "RAD21, SYCP1, SYCP2, SYCP3, SYCE1, SYCE2, SYCE3, HORMAD1, HORMAD2, MND1". The list of tissues includes:

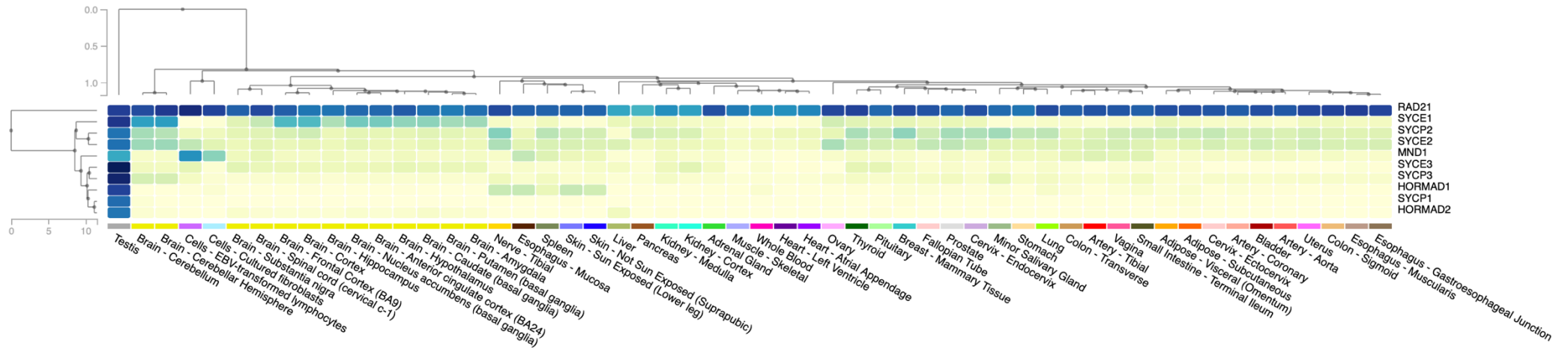
- Brain
- Brain - Amygdala
- Brain - Anterior cingulate cortex (BA24)
- Brain - Caudate (basal ganglia)
- Brain - Cerebellar Hemisphere
- Brain - Cerebellum
- Brain - Cortex
- Brain - Frontal Cortex (BA9)
- Brain - Hippocampus
- Brain - Hypothalamus
- Brain - Nucleus accumbens (basal ganglia)
- Brain - Putamen (basal ganglia)
- Brain - Spinal cord (cervical c-1)
- Brain - Substantia nigra
- Adipose Tissue
- Adipose - Subcutaneous
- Adipose - Visceral (Omentum)
- Adrenal Gland
- Bladder
- Blood Vessel
- Artery - Aorta
- Artery - Coronary
- Artery - Tibial
- Cells - EBV-transformed lymphocytes
- Breast - Mammary Tissue
- Cervix Uteri
- Cervix - Ectocervix
- Cervix - Endocervix
- Colon
- Colon - Sigmoid
- Colon - Transverse
- Esophagus
- Esophagus - Gastroesophageal Junction
- Esophagus - Mucosa
- Esophagus - Muscularis
- Fallopian Tube
- Heart
- Heart - Atrial Appendage
- Heart - Left Ventricle
- Kidney
- Kidney - Cortex
- Kidney - Medulla
- Liver
- Lung
- Minor Salivary Gland
- Muscle - Skeletal
- Nerve - Tibial
- Ovary
- Pancreas
- Pituitary
- Prostate
- Skin
- Cells - Cultured fibroblasts
- Skin - Not Sun Exposed (Suprapubic)
- Skin - Sun Exposed (Lower leg)
- Small Intestine - Terminal Ileum
- Spleen
- Stomach
- Testis
- Thyroid
- Uterus
- Vagina
- Whole Blood

GTEx Portal Examples

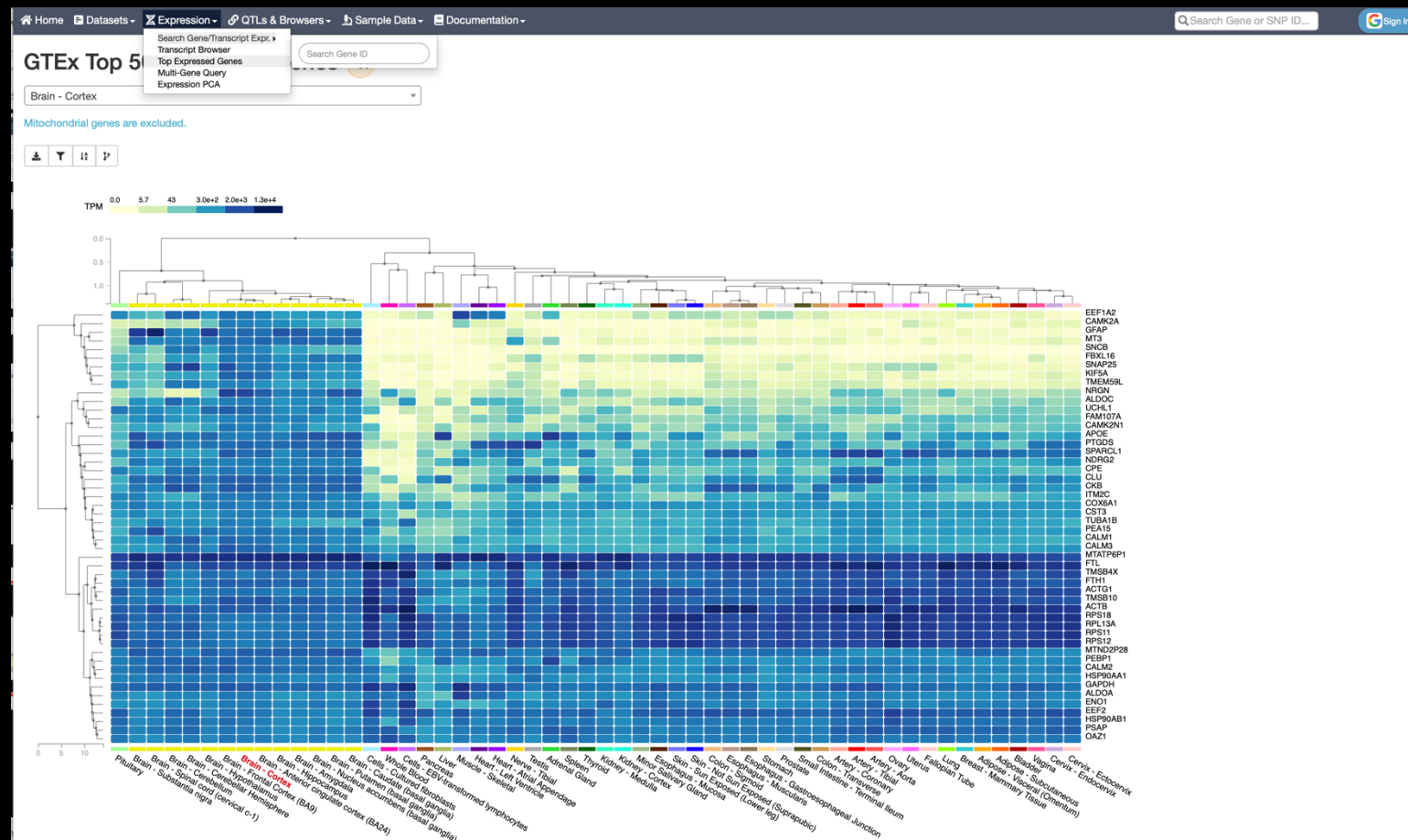
GTEx Multi Gene Query



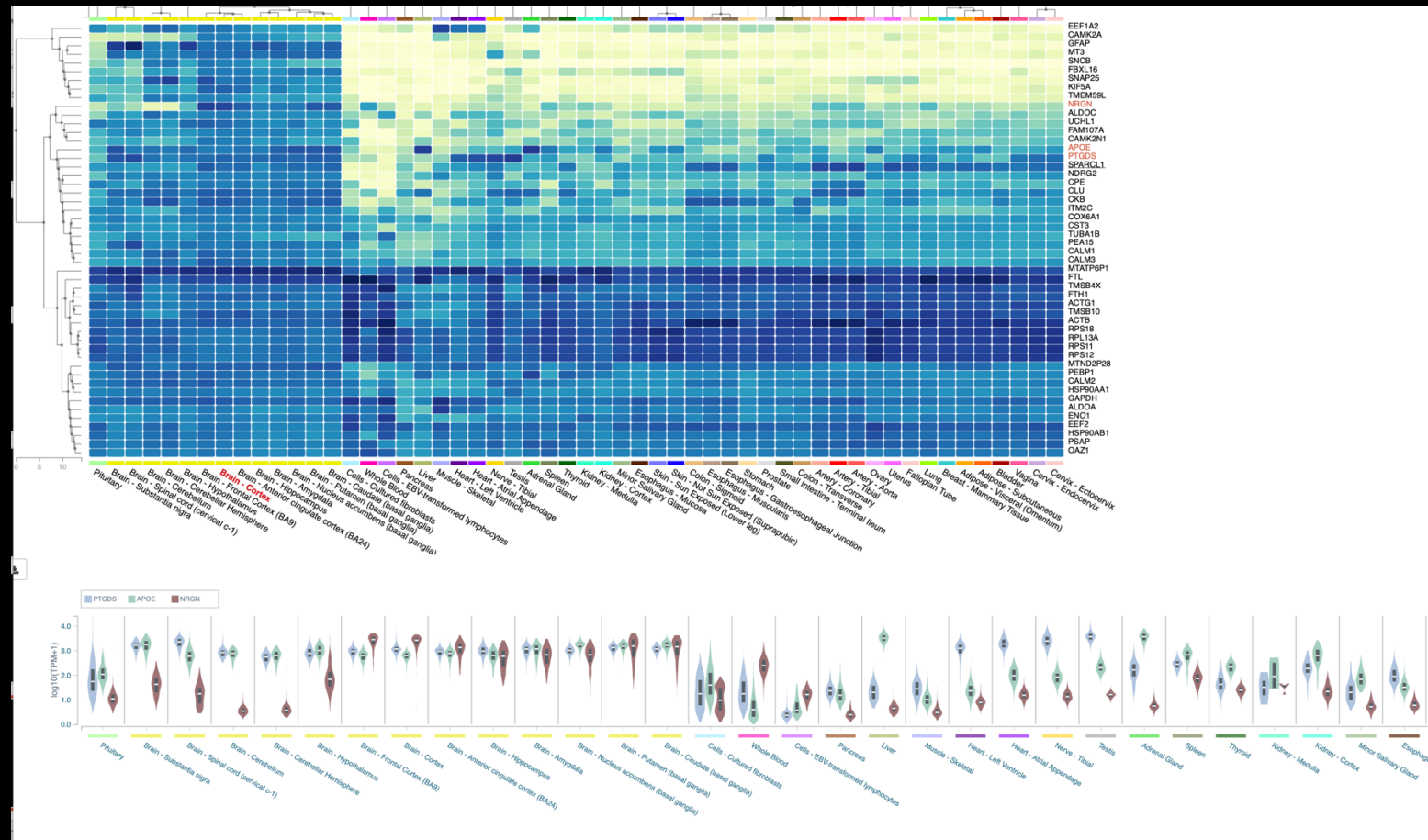
TPM 0.0 2.0 7.7 25 75 2.2e+2



Search top expressed genes by tissue

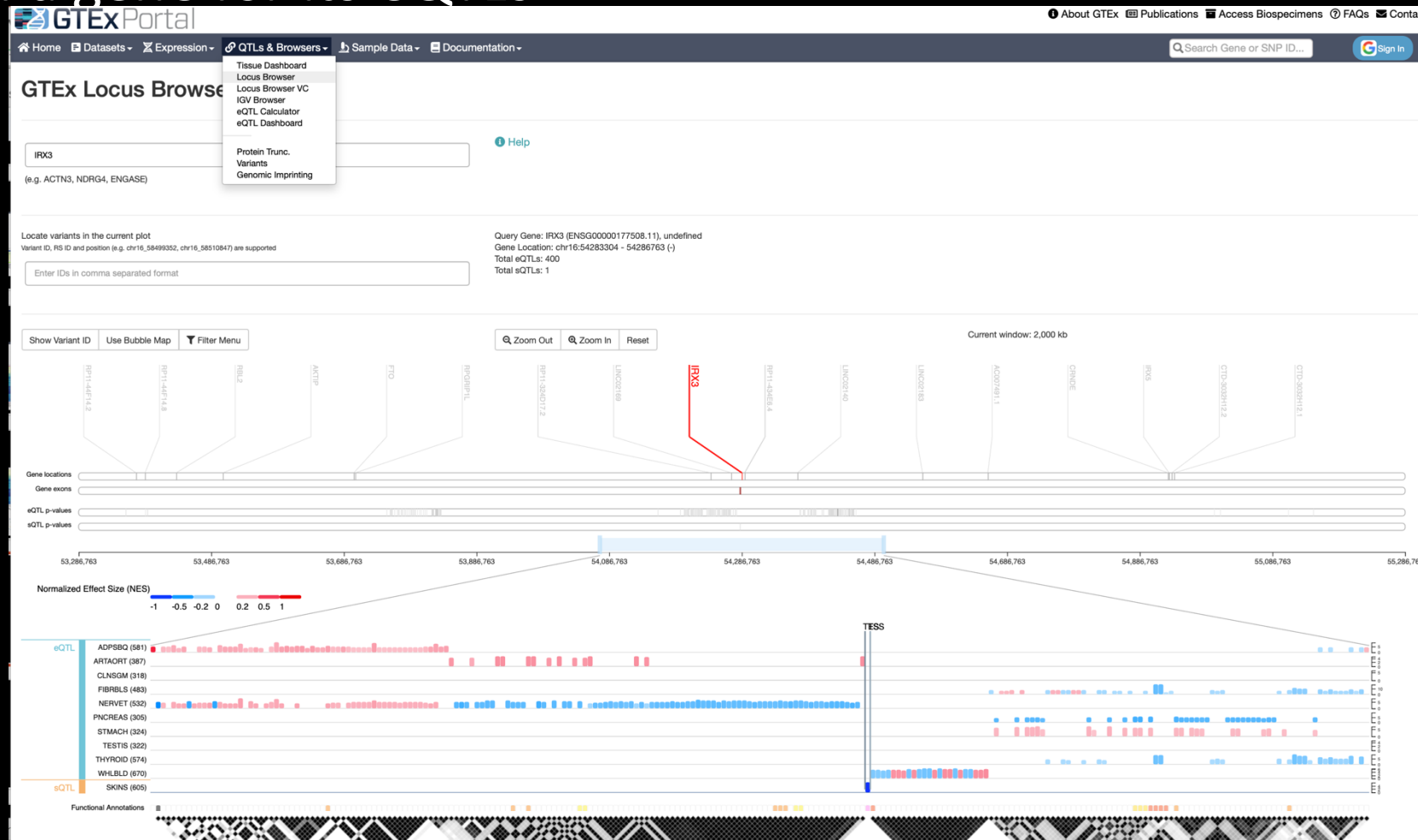


Select genes to show violin plots

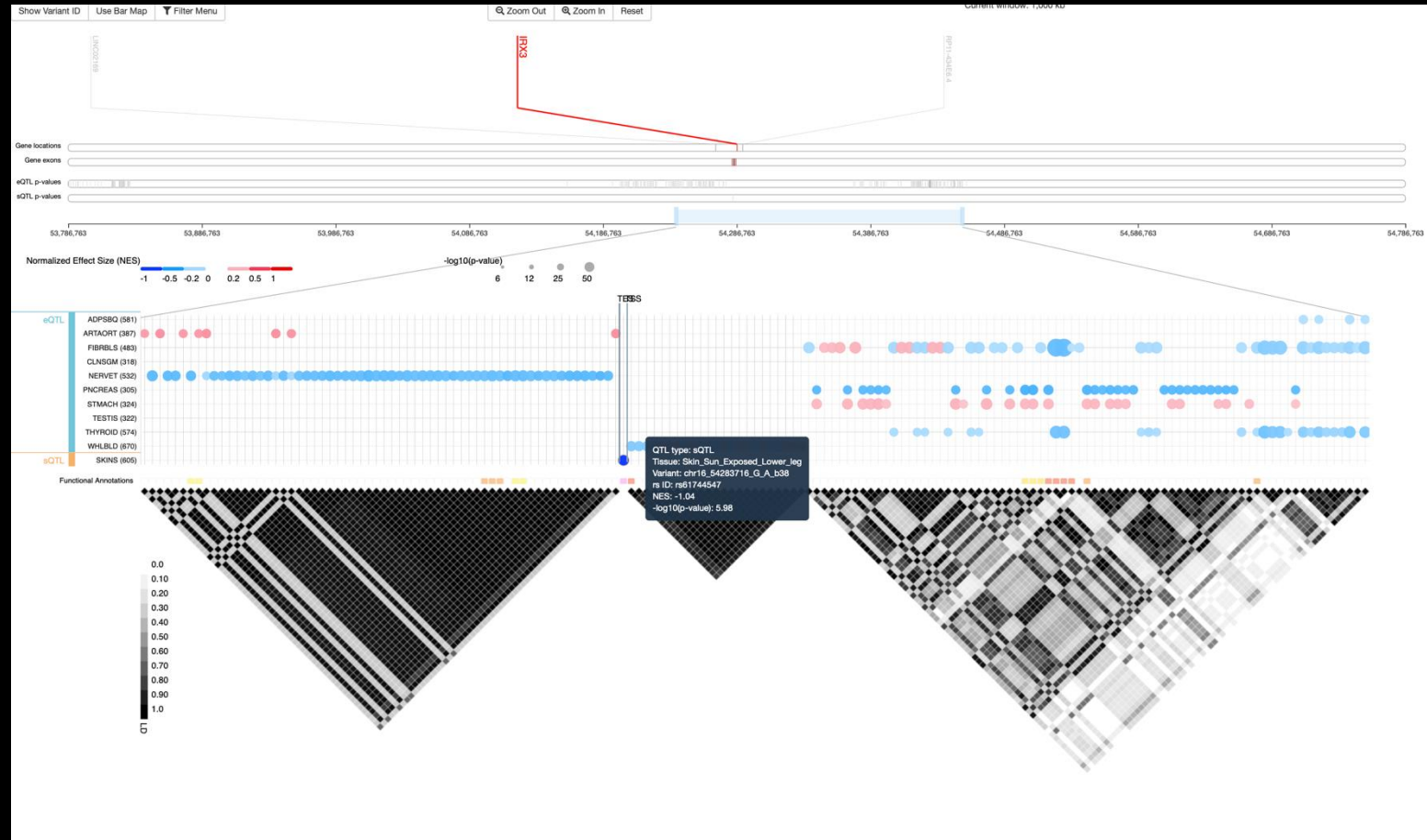


GTEx Portal Examples

Search a gene for its eQTLs



GTEx Portal Examples



Genotype + transcriptome integration

Earlier we talked about reasons for testing *genes* instead of only SNPs.

We asked:

- Can we quantify/define “gene function” in a simple/systematic way?
- Can we actually measure this for individuals somehow?
- Can we then test each gene’s “gene function” for association with a trait?

Genotype + transcriptome integration

Earlier we talked about reasons and methods for testing *genes* instead of only SNPs.

- Can we quantify/define “gene function” in a simple/systematic way?
Yes, by considering expression as an intermediate variable between genotype and phenotype
- Can we actually measure this for individuals somehow?
Yes, by using eQTL databases to impute individuals’ expression (for each gene/tissue) from their genotypes
- Can we then test each gene’s “gene function” for association with a trait?
Yes, by performing transcriptome-wide association study (TWAS)

TWAS

Input data

- GWAS dataset
 - Genotypes
 - Phenotypes
- Transcriptome database
 - eQTL associations as a “training set” for predicting/imputing transcription in your GWAS data

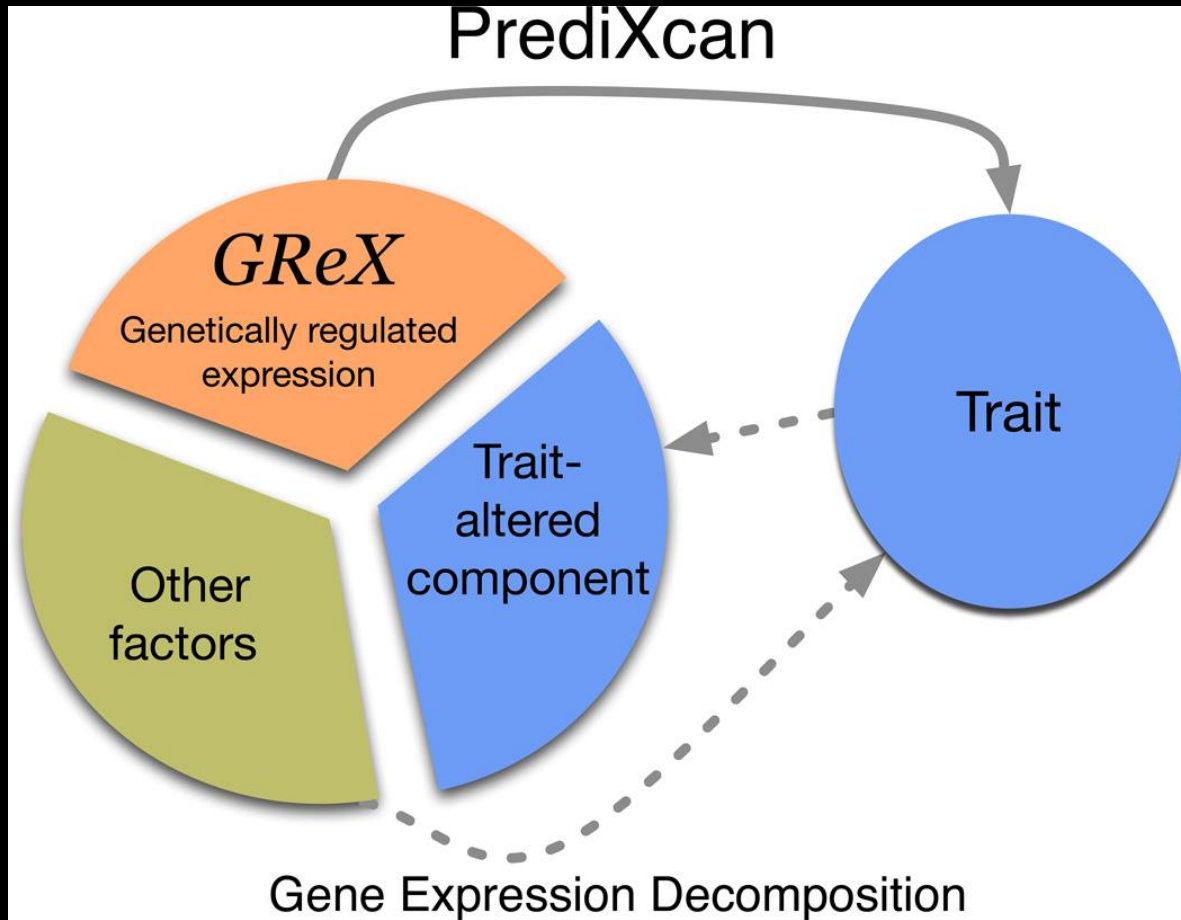
TWAS

There are different methods for implementing the TWAS idea.
Here we'll focus on PrediXcan.

See:

Gamazon, E. R., Wheeler, H. E., Shah, K. P., Mozaffari, S. V., Aquino-Michaels, K., Carroll, R. J., Eyler, A. E., Denny, J. C., GTEx Consortium, Nicolae, D. L., Cox, N. J., & Im, H. K. (2015). A gene-based association method for mapping traits using reference transcriptome data. *Nature genetics*, 47(9), 1091–1098. <https://doi.org/10.1038/ng.3367>
(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4552594/>)

PrediXcan conceptual model

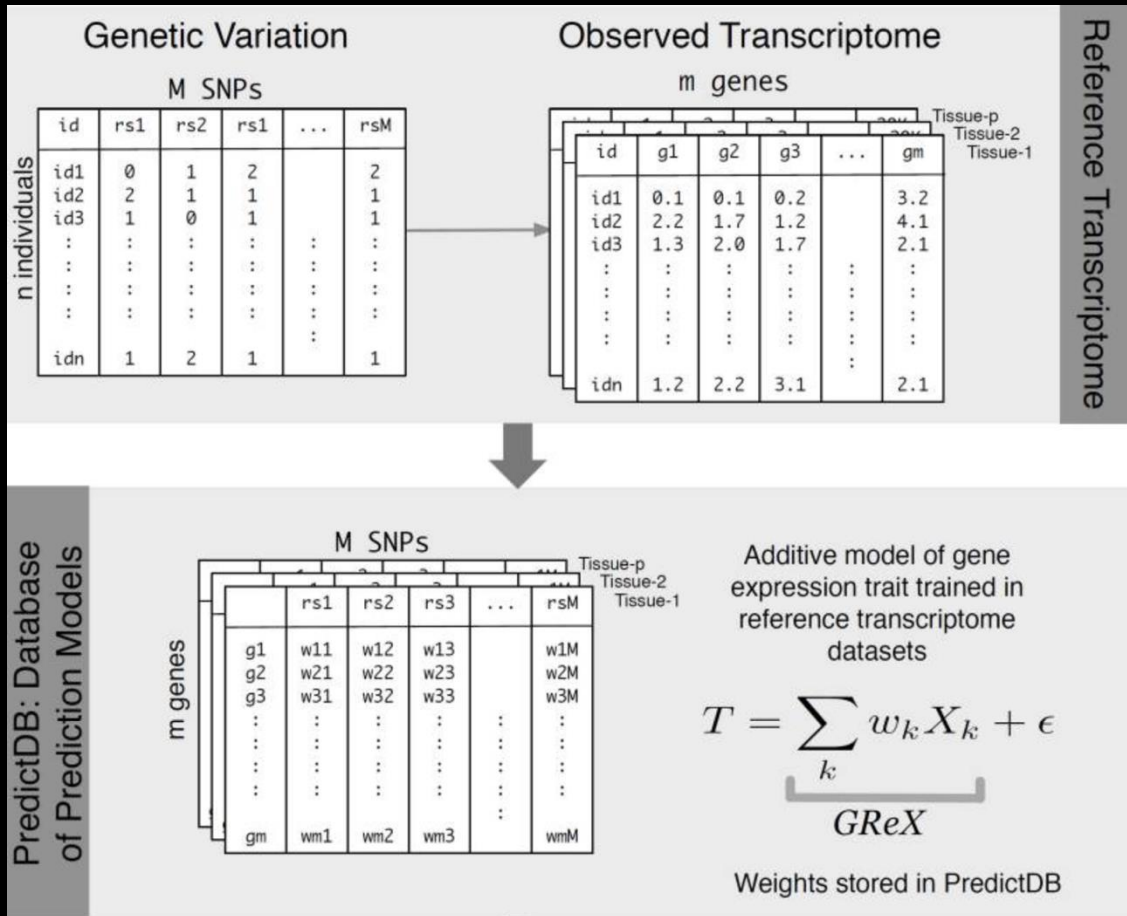


Mechanism tested by the PrediXcan method

This figure shows the conceptual decomposition of the expression level of a gene into three components: genetically determined component, a component altered by the trait itself, and the remaining factors (including environment). PrediXcan estimates the genetically regulated component of expression (*GReX*) and correlates it with the trait to identify trait-associated genes.

We denote the estimated value with a hat, $\widehat{GReX}$

PrediXcan – training data



Top panel: Reference data (a transcriptome database like GTEx). Known genotypes + known transcription levels in several tissues.

Bottom panel: Train a model for using genotype to predict expression of each gene in each tissue.

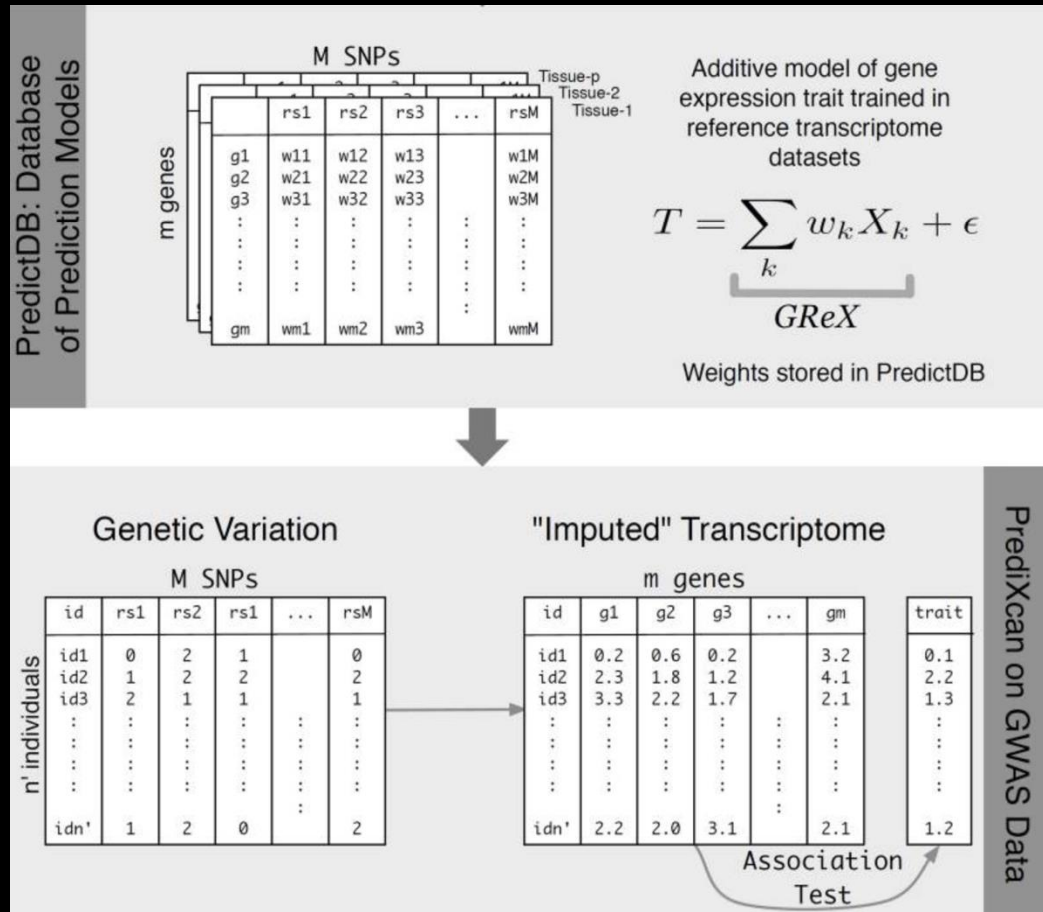
Model essentially involves choosing which variants (eQTLs) to use as predictors and what “weight” (effect) to assign for each variant’s genotype. Weights are “learned” using statistical learning algorithms such as LASSO or elastic net.

T = transcription level of a given gene

X_k = number of ref. alleles (0, 1, or 2) for the k^{th} SNP being used to predict T for the gene

w_k = effect for k^{th} SNP

PrediXcan - TWAS



Top panel (same as bottom panel from previous slide):
Train a model for using genotype to predict expression of each gene in each tissue.

Bottom panel:
Apply the model to your GWAS data to calculate estimated transcription of each gene in tissues of interest.
Now test each gene's transcription for association with trait.

Can use linear/logistic/Cox regression, non-parametric approaches, etc., as appropriate.

PrediXcan – note on model selection

In the model, a variety of SNP selection techniques are possible

- Using the top SNP (strongest eQTL) only
- Use the top few SNPs (e.g., eQTLs with $p < 1E-2$)
- Use *all* the cis-eQTLs
- Use penalized regression methods to select only *some* SNPs
 - LASSO forces all but a few of the weights to be 0
 - Elastic net shrinks some of the weights without forcing them to be 0
- Include trans-eQTLs, too?

Elastic net and LASSO perform best; authors preferred elastic net.

Including trans-eQTLs didn't seem to improve prediction much.

PrediXcan – note on model selection

How to decide what model is best?

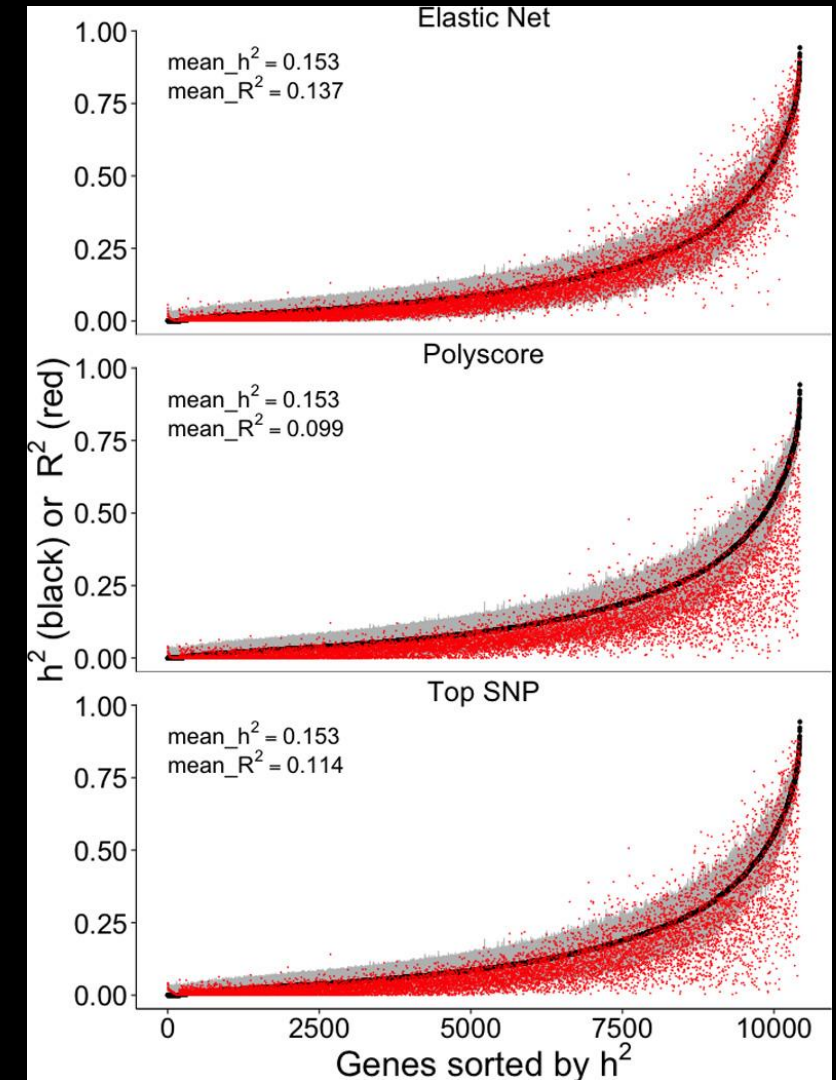
The plot shows performance in the training data.

Out-of-sample testing was done, too.

Also looked at observed vs. predicted expression for specific genes.

Cross-validated prediction performance vs heritability

This figure shows the prediction performance (R^2 of GReX vs. observed expression in red) compared to gene expression heritability estimates (black with 95% confidence interval in gray). Performance was assessed using 10-fold cross-validation in the DGN whole blood cohort ($n=922$) with the elastic net, polygenic score ($p < 1 \times 10^{-4}$), and using the top SNP for prediction.



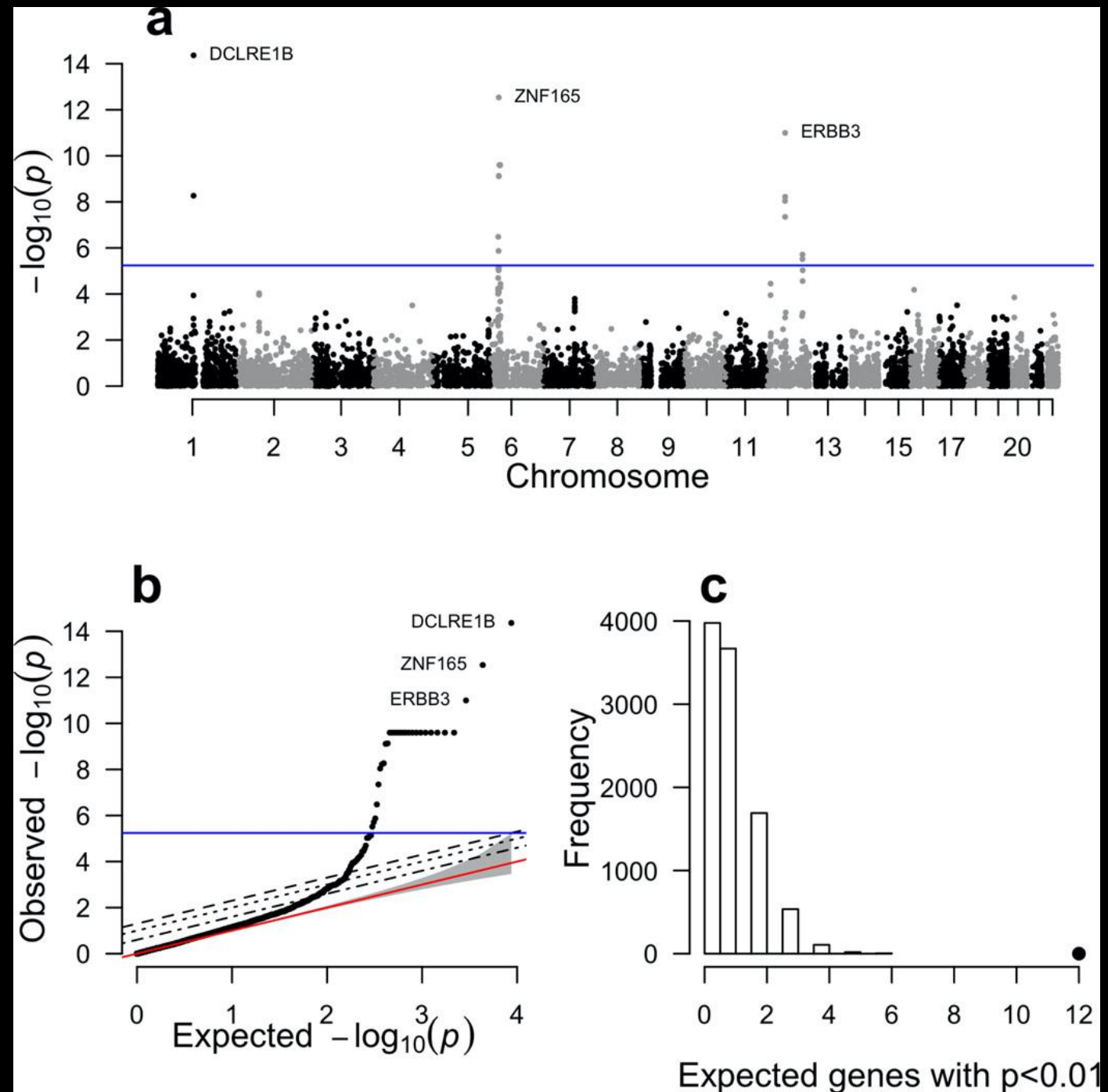
Applying PrediXcan

PrediXcan was run using whole blood + elastic net models for several traits (T1D shown).

Reasonably good at finding associations with previously known T1D genes.

At least some replication when applied to different GWAS data for the same trait

Also, perform gene set analyses for “top hit” genes



Running PrediXcan

PrediXcan is freely available as a pair of command-line Python programs (MetaXcan is also required).

Various models have been precomputed:

<http://predictdb.org>

Tutorial/manual:

<https://github.com/hakyimlab/MetaXcan/wiki/Individual-level-PrediXcan:-introduction,-tutorials-and-manual>

Running PrediXcan

Example 1: VCF hg19-based genotype, on GTEx v8 Elastic Net Model

The easiest example is one where the genotypes and models use the same variants. Here we show how to run on Thousand Genomes genotypes from hg37 with the Elastic Net models:

```
printf "Predict expression\n\n"
python3 $METAXCAN/Predict.py \
--model_db_path $DATA/models/gtex_v8_en/en_Whole_Blood.db \
--vcf_genotypes $DATA/1000G_hg37/ALL.chr*.phase3_shapeit2_mvncall_integrated_v5a.20130502.genotypes.vcf.gz
--vcf_mode genotyped \
--prediction_output $RESULTS/vcf_1000G_hg37_en/Whole_Blood__predict.txt \
--prediction_summary_output $RESULTS/vcf_1000G_hg37_en/Whole_Blood__summary.txt \
--verbosity 9 \
--throw

printf "association\n\n"
python3 $METAXCAN/PrediXcanAssociation.py \
--expression_file $RESULTS/vcf_1000G_hg37_en/Whole_Blood__predict.txt \
--input_phenos_file $DATA/1000G_hg37/random_pheno_1000G_hg37.txt \
--input_phenos_column pheno \
--output $RESULTS/vcf_1000G_hg37_en/Whole_Blood__association.txt \
--verbosity 9 \
--throw
```

Note: “mapping” your GWAS variants to the variant IDs in the pre-calculated model may take some work

TWAS with *only* GWAS summary statistics

Earlier we said:

Input data

- GWAS dataset
 - Genotypes
 - Phenotypes
- Transcriptome database
 - eQTL associations as a “training set” for predicting/imputing transcription in your GWAS data

TWAS with *only* GWAS summary statistics

What if you only have

- ~~GWAS dataset~~

- ~~Genotypes~~

- ~~Phenotypes~~

- GWAS summary statistics

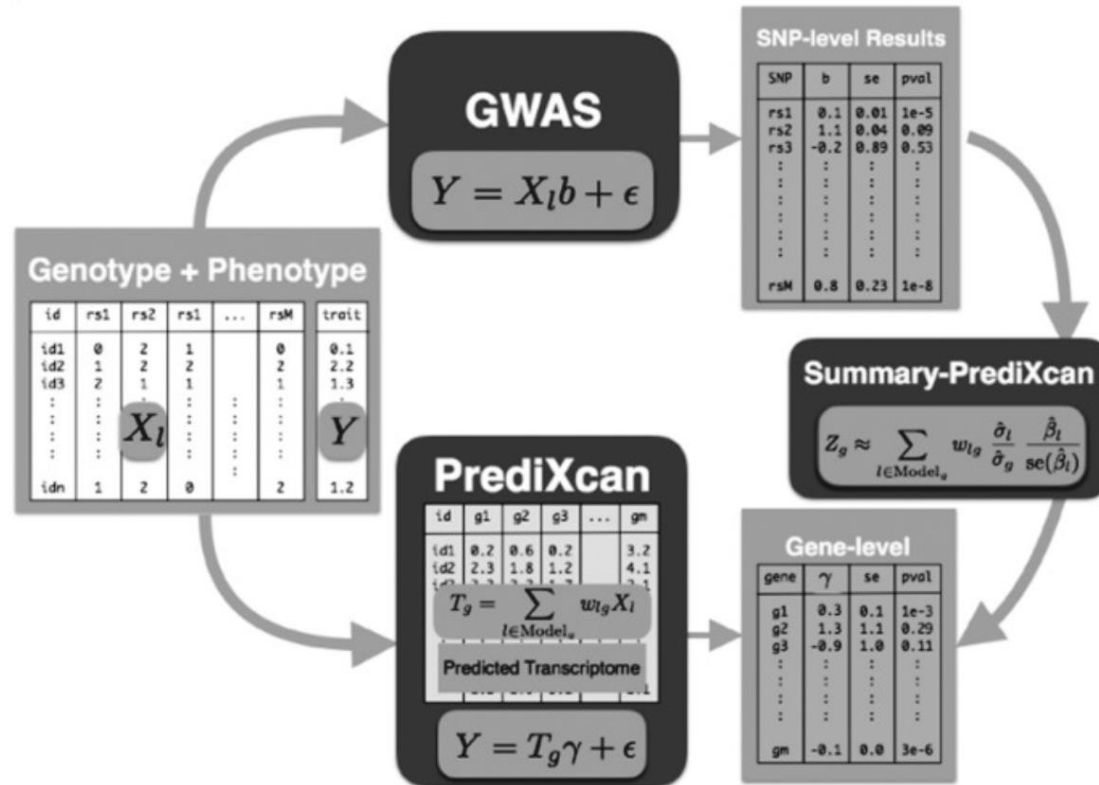
- Transcriptome database

- eQTL associations as a “training set” for predicting/imputing transcription in your GWAS data

Summary statistic TWAS with S-PrediXcan

Fig. 1

a



Barbeira, A. N., Dickinson, S. P., Bonazzola, R., Zheng, J., Wheeler, H. E., Torres, J. M., Torstenson, E. S., Shah, K. P., Garcia, T., Edwards, T. L., Stahl, E. A., Huckins, L. M., GTEx Consortium, Nicolae, D. L., Cox, N. J., & Im, H. K. (2018). Exploring the phenotypic consequences of tissue specific gene expression variation inferred from GWAS summary statistics. *Nature communications*, 9(1), 1825.

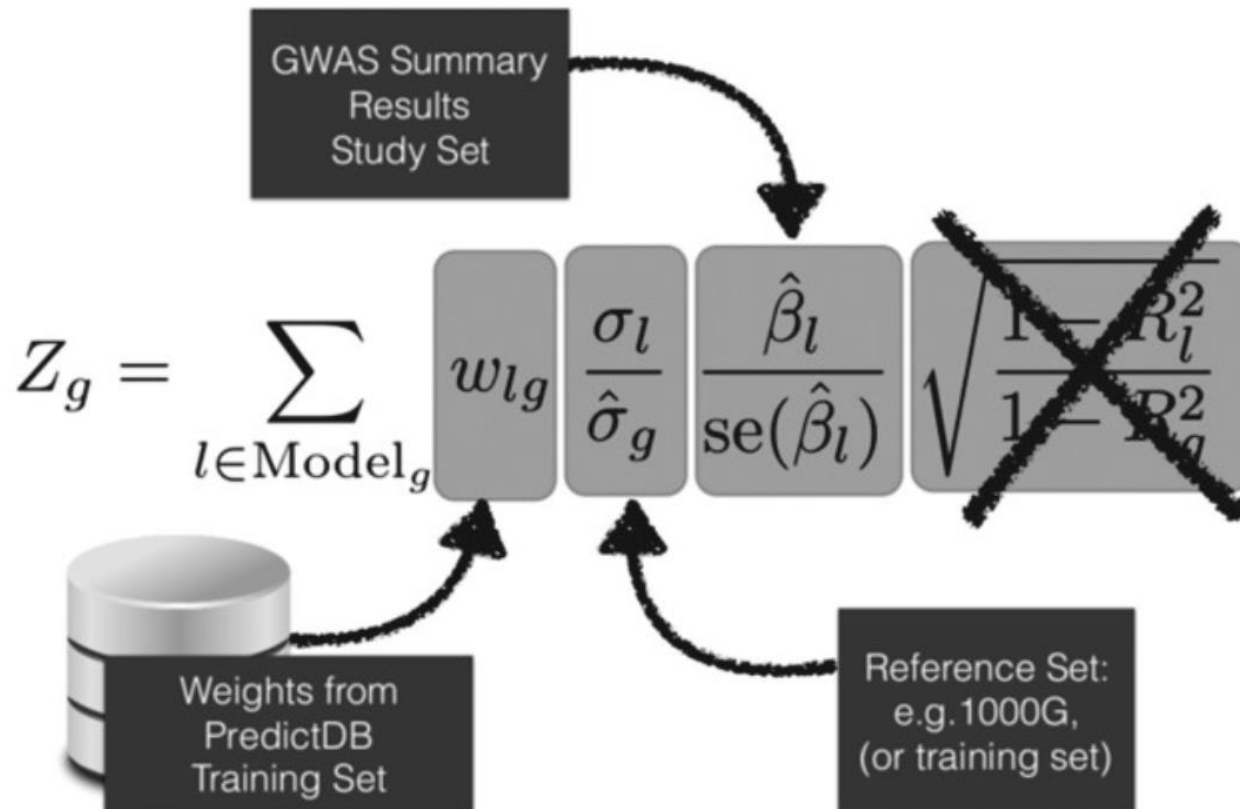
<https://doi.org/10.1038/s41467-018-03621-1>

<https://pubmed.ncbi.nlm.nih.gov/29739930/>

S-PrediXcan

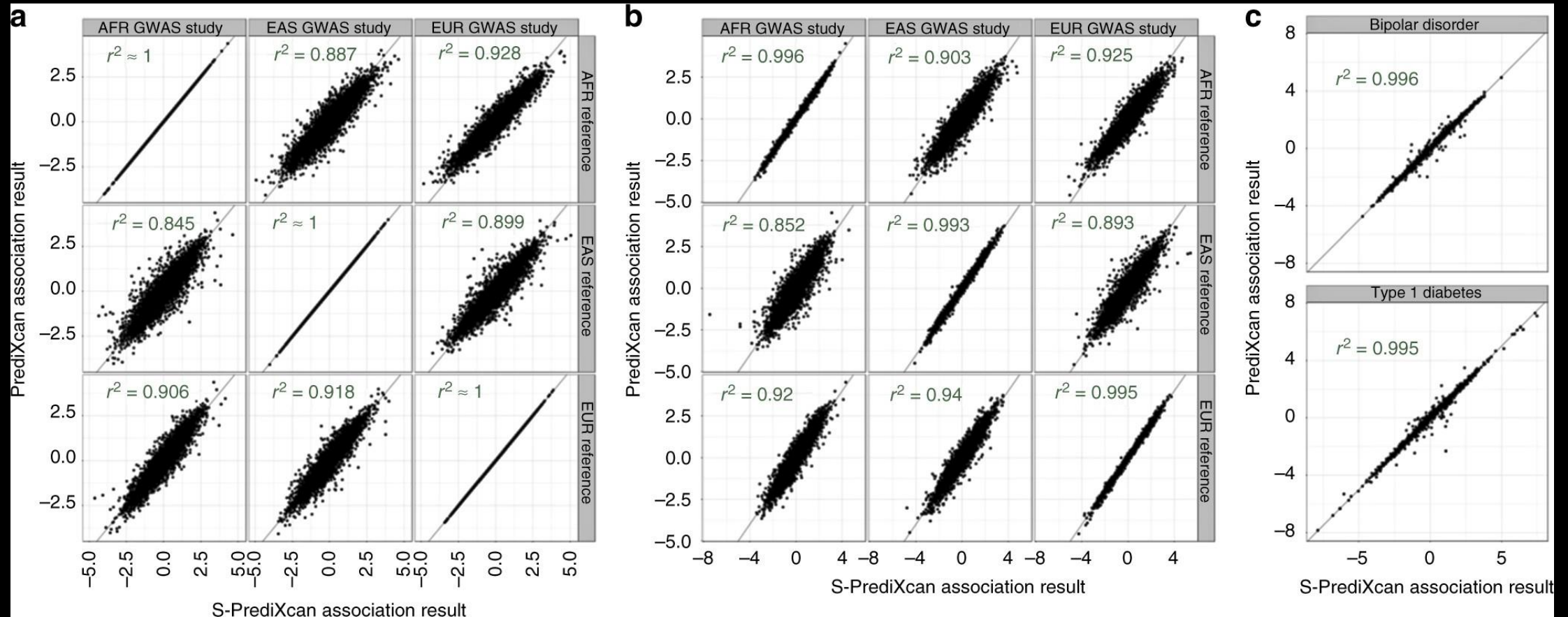
“We find high concordance between PrediXcan and S-PrediXcan results indicating that in most cases, we can use the summary version without loss of power to detect associations.”

b



b Shows the components of the formula to calculate PrediXcan gene-level association results using summary statistics. The different sets involved as input data are shown. The regression coefficient between the phenotype and the genotype is obtained from the study set. The training set is the reference transcriptome dataset where the prediction models of gene expression levels are trained. The reference set (1000G, or training set having some advantages) is used to compute the variances and covariances (LD structure) of the markers used in the predicted expression levels. Both the reference set and training set values are precomputed and provided to the user so that only the study set results need to be provided to the software. The crossed out term was set to 1 as an approximation. We found this approximation to have negligible impact on the results

S-PrediXcan performance



Comparison of PrediXcan and S-PrediXcan results in real and simulated traits. This figure shows a comparison of PrediXcan vs. S-PrediXcan for **a** a simulated phenotype under null hypothesis of no genetic component; **b** a cellular phenotype (=intrinsic growth); and **c** bipolar disorder and type 1 diabetes studies from Wellcome Trust Case Control Consortium (WTCCC). Gene expression prediction models were based on the DGN cohort presented in ref. ¹¹. For the simulated phenotype, study sets (GWAS set) and reference sets (LD calculation set) consisted of African (661), East Asian (504), and European (503) individuals from the 1000 Genomes Project. When the same study set is used as reference set, we obtained a high correlation (coefficient of determination): $r^2 > 0.99999$. For the intrinsic growth phenotype, study sets were a subset of 140 individuals from each of the African, Asian, and European groups from 1000 Genomes Project. The reference set was the same as for the simulated phenotype. For the disease phenotypes, the study set consisted of British individuals, and the LD calculation set was the European population subset of the 1000 Genomes Project

MetaXcan: S-PrediXcan “pipeline”/framework

S-PrediXcan combines GWAS signals and eQTL signals based on linkage disequilibrium, and it outputs p-values relating gene expression to the GWAS trait.

If a gene is significant in S-PrediXcan, it could possibly be the result of “LD contamination” due to nearby but independent eQTL/GWAS signals.

In the suggested pipeline, S-PrediXcan results should be filtered to exclude associations based on “non-colocalized” signals.

The authors have made available data/tools for this.

<http://gene2pheno.org>

<https://github.com/hakyimlab/phenomexcan>

S-PrediXcan/MetaXcan implementation

S-PrediXcan/MetaXcan is implemented in Python
<https://github.com/hakyimlab/MetaXcan>

S-PrediXcan Input data

S-PrediXcan will calculate the gene-level association results from GWAS summary statistics. It supports most GWAS formats by accepting command line argument specifying data columns. Some precalculated data is needed, that must be set up prior to S-PrediXcan execution.

The gist of S-PrediXcan's input is:

- A Transcriptome Prediction Model database (an example is [here](#))
- A file with the covariance matrices of the SNPs within each gene model (such as [this one](#))
- GWAS results (such as [these](#), which were computed on a randomly generated phenotype). GWAS results can belong to a single file or be split into multiple ones (i.e. split by chromosome). You can specify the necessary columns via command line arguments (i.e. which column holds snps, which holds p-values, etc)

You can use precalculated databases, or generate new ones with tools available in [PredictDB repository](#). GTEx-based tissues and 1000 Genomes covariances precalculated data can be found [here](#).

S-PrediXcan/MetaXcan implementation

```
./SPrediXcan.py \  
--model_db_path data/DGN-WB_0.5.db \  
--covariance data/covariance.DGN-WB_0.5.txt.gz \  
--gwas_folder data/GWAS \  
--gwas_file_pattern "*.gz" \  
--snp_column SNP \  
--effect_allele_column A1 \  
--non_effect_allele_column A2 \  
--beta_column BETA \  
--pvalue_column P \  
--output_file results/test.csv
```

Further extension: (S-)MultiXcan

“We develop a new method, MultiXcan, to test the mediating role of gene expression variation on complex traits, integrating information available across multiple tissue studies. We show this approach has higher power than traditional single-tissue methods. We extend this method to use only summary-statistics from public GWAS.”

Barbeira, A. N., Pividori, M., Zheng, J., Wheeler, H. E., Nicolae, D. L., & Im, H. K. (2019). Integrating predicted transcriptome from multiple tissues improves association detection. *PLoS genetics*, 15(1), e1007889.

<https://doi.org/10.1371/journal.pgen.1007889>

<https://pubmed.ncbi.nlm.nih.gov/30668570/>

“MultiXcan (**MuTiXcan.py**) and S-MultiXcan (**SMuTiXcan.py**) compute omic associations, integrating measurements across tissues while factoring correlation. For example, if you have prediction models, each trained on different regions of the brain, MuTiXcan will combine the information across all experiments. This is effectively a meta-analysis across tissues, where each tissue is an experiment and we explicitly account for correlation.”

<https://github.com/hakyimlab/MetaXcan>

Further evaluation of PrediXcan

“We tested the prediction accuracy of PrediXcan by comparing the predicted and observed gene expression levels, and also looked into some potential influential factors and a filter criterion with the aim of improving PrediXcan performance.”

Li, B., Verma, S. S., Veturi, Y. C., Verma, A., Bradford, Y., Haas, D. W., & Ritchie, M. D. (2018). Evaluation of PrediXcan for prioritizing GWAS associations and predicting gene expression. *Pacific Symposium on Biocomputing*. *Pacific Symposium on Biocomputing*, 23, 448–459.

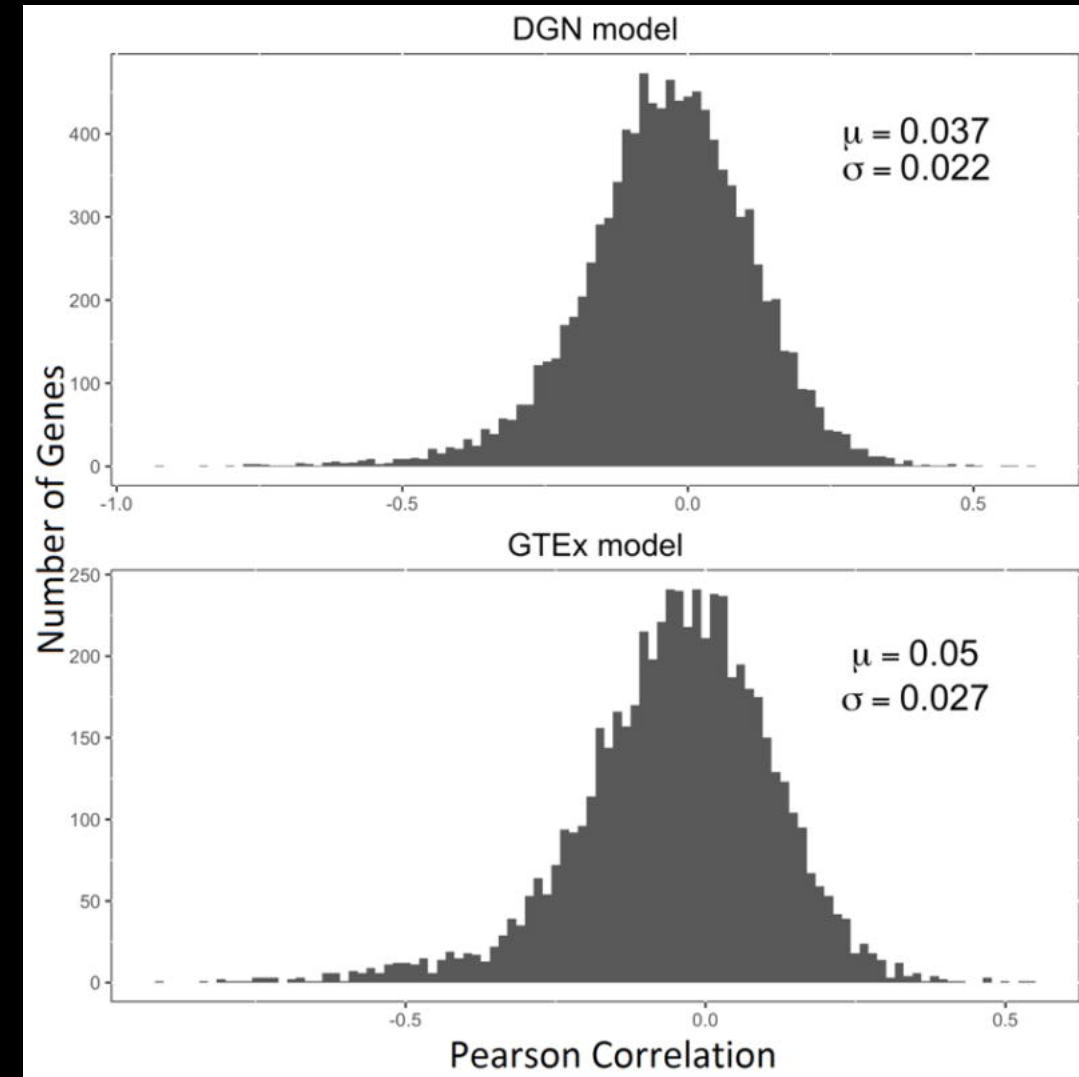
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5749400/>

Further evaluation of PrediXcan

“prediction accuracy was overall unsatisfactory as most genes had accuracy estimates near 0”

“For the two whole blood models, the directionality estimates centered on zero with a small standard deviation, which suggested that most predicted gene expression levels did not correlate with the observed gene expression levels ”

Authors suggest filtering to enrich for genes that have similar predictions for the same tissue based on models built from two or more databases



Performance of prediction directionality of PrediXcan models, DGN (top) and GTEx (bottom), on the YRI cohort. Directionality was computed between predicted and observed gene expression levels.

Further evaluation of PrediXcan

How useful is PrediXcan for prioritizing association results (from GWAS, e.g.)?

“When it came to prioritizing genetic association study results, PrediXcan was able to identify genes that were not significant in GWAS, and also retained significant variant-trait associations. These results were reassuring of the utility of PrediXcan. PrediXcan possessed promising features to reduce research burden by focusing on genes instead of SNPs, and map regulatory effects of distant SNPs onto responding genes, which are overlooked by most studies where only genes adjacent to SNPs are investigated[...] Overall, the present study found that PrediXcan performed differently when evaluated for different functions.”

How accurate is PrediXcan across populations?

7 PrediXcan weight databases

- DGN whole-blood (further referred to as DGN)
- GTEx v6 1KG whole blood
- GTEx v6 1KG LCL
- GTEx v6 HapMap whole blood
- GTEx v6 HapMap LCL, GTEx v7
- HapMap whole blood (GTEx WB)
- GTEx v7 HapMap LCL (GTEx LCL)

Gene expression data: RNA-seq on lymphoblastoid cell line (LCL) samples for 464 individuals from five populations

Of these, 445 subjects were in the 1000 Genomes Phase 3 dataset

- 358 subjects of European descent
 - Utah residents with Northern and Western European ancestry (CEU, $n = 89$)
 - British individuals in England and Scotland (GBR, $n = 86$)
 - Finnish in Finland (FIN, $n = 92$)
 - Toscani in Italy (TSI, $n = 91$).
- 87 individuals of African descent from Yoruba in Ibadan, Nigeria (YRI, $n = 87$)

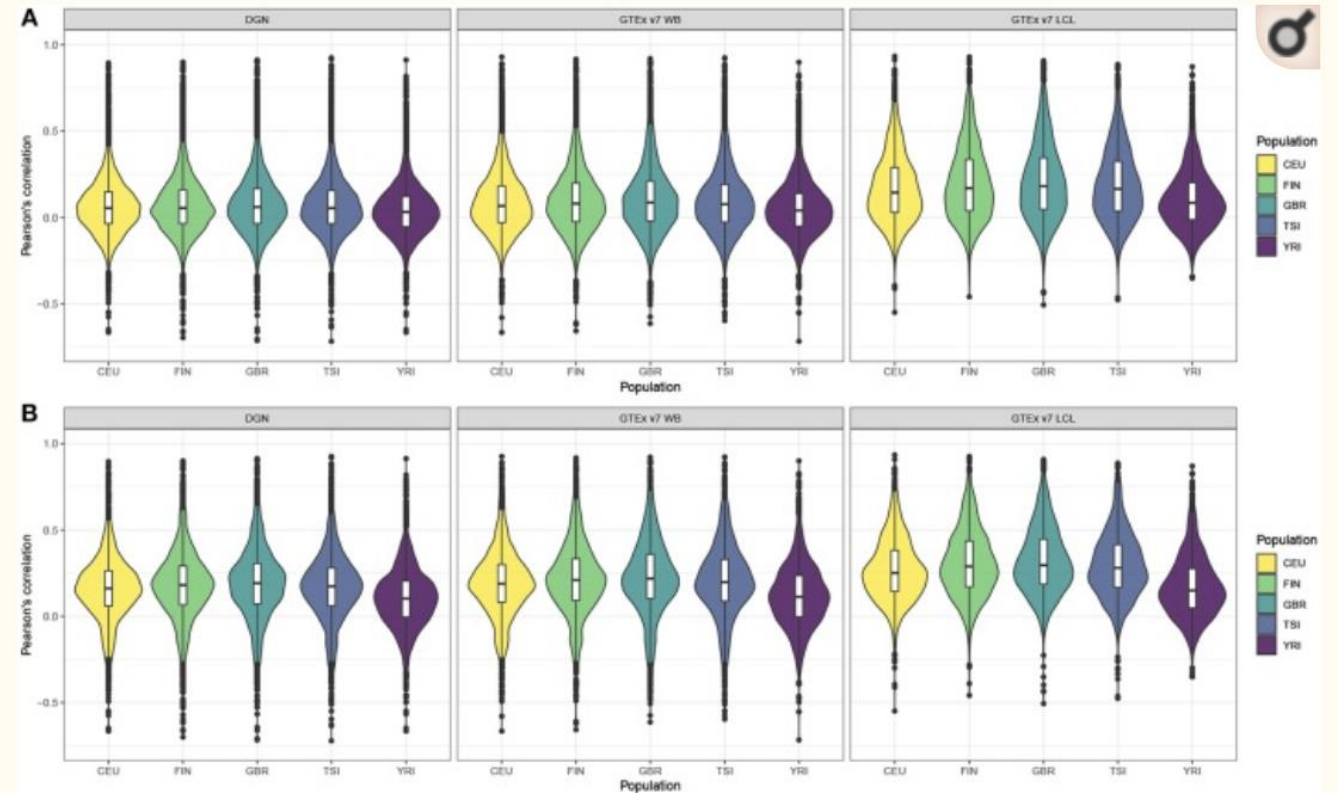
Mikhaylova, A. V., & Thornton, T. A. (2019). Accuracy of Gene Expression Prediction From Genotype Data With PrediXcan Varies Across and Within Continental Populations. *Frontiers in genetics*, 10, 261.

<https://doi.org/10.3389/fgene.2019.00261>

<https://pubmed.ncbi.nlm.nih.gov/31001318/>

How accurate is PrediXcan across populations?

The majority of genes have very poor prediction accuracy—of the genes predicted with whole-blood databases, a third have negative correlations and a half have correlations between 0 and 0.2. Of the genes predicted with LCL, a fifth have negative correlations and over a third have correlations between 0 and 0.2. [...]. The predictive accuracy is the lowest in the African sample. Across all populations, only a small number of genes were predicted with high accuracy (with $r > 0.6$). [...]. Surprisingly, we also find highly significant differences in prediction accuracy with PrediXcan among the closely related European ancestry populations, with the Finnish, British, and Italian populations having significantly higher prediction accuracy than the CEU. These results are consistent across all seven PrediXcan weight databases we considered.



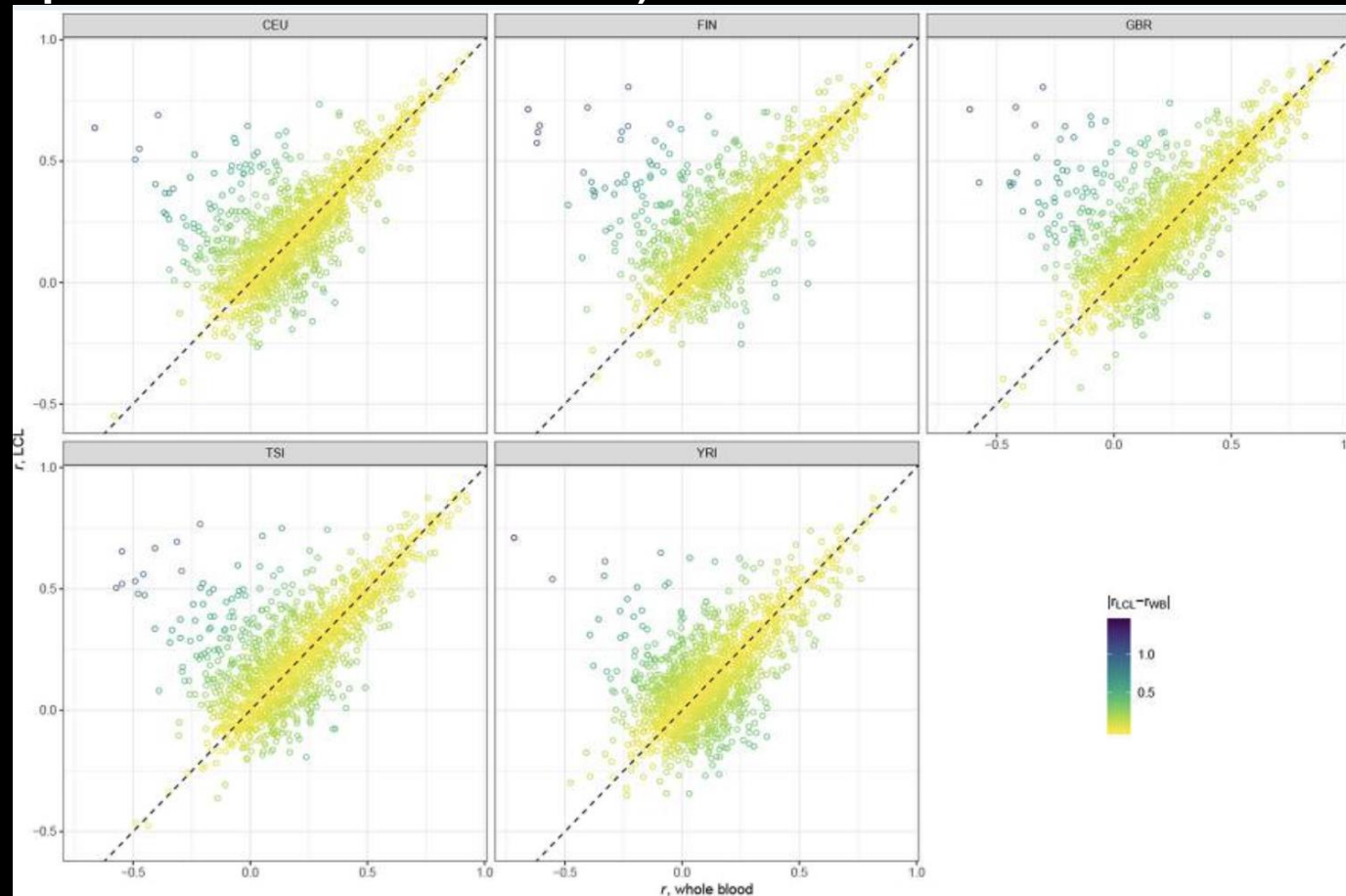
[Open in a separate window](#)

Figure 1

Violin plots of gene expression correlation coefficients by five populations using DGN, GTEx v7 WB, and GTEx v7 LCL weight databases; (A) before and (B) after filtering out poorly predicted genes.

How accurate is PrediXcan across populations?

**PrediXcan Prediction Accuracy Differs Between Tissues
(lymphoblastoid > whole blood)**



Further suggested reading

Perspective | Published: 29 March 2019

Opportunities and challenges for transcriptome-wide association studies

Michael Wainberg, Nasa Sinnott-Armstrong, Nicholas Mancuso, Alvaro N. Barbeira, David A. Knowles, David Golan, Raili Ermel, Arno Ruusalepp, Thomas Quertermous, Ke Hao, Johan L. M. Björkegren , Hae Kyung Im , Bogdan Pasaniuc , Manuel A. Rivas  & Anshul Kundaje 

Nature Genetics **51**, 592–599(2019) | [Cite this article](#)

18k Accesses | **113** Citations | **71** Altmetric | [Metrics](#)

Abstract

Transcriptome-wide association studies (TWAS) integrate genome-wide association studies (GWAS) and gene expression datasets to identify gene–trait associations. In this Perspective, we explore properties of TWAS as a potential approach to prioritize causal genes at GWAS loci, by using simulations and case studies of literature-curated candidate causal genes for schizophrenia, low-density-lipoprotein cholesterol and Crohn’s disease. We explore risk loci where TWAS accurately prioritizes the likely causal gene as well as loci where TWAS prioritizes multiple genes, some likely to be non-causal, owing to sharing of expression quantitative trait loci (eQTL). TWAS is especially prone to spurious prioritization with expression data from non-trait-related tissues or cell types, owing to substantial cross-cell-type variation in expression levels and eQTL strengths. Nonetheless, TWAS prioritizes candidate causal genes more accurately than simple baselines. We suggest best practices for causal-gene prioritization with TWAS and discuss future opportunities for improvement. Our results showcase the strengths and limitations of using eQTL datasets to determine causal genes at GWAS loci.

<https://www.nature.com/articles/s41588-019-0385-z>

Another common TWAS framework: FUSION

<http://gusevlab.org/projects/fusion/>

<https://www.nature.com/articles/ng.3506>